

Kola Nikhil Jai

Hyderabad, Telangana, India

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Experience

CDPL | *Software Engineering Intern*

Feb 2026 - Present

- Developed and deployed a full-stack web platform using Next.js, hosted on Vercel with custom domain mapping and continuous UI/UX enhancements
- Built scalable backend services and optimized APIs, improving performance, reliability, and system responsiveness under real-world usage
- Leveraged AWS S3 for efficient media storage and delivery, significantly reducing load times and improving website performance
- Engineered a kiosk-based drone simulation interface using Python Tkinter, enabling real-time interaction and immersive system visualization
- Exploring integration of LLM-based intelligence into drone systems for real-time decision-making, automation, and adaptive control

Personal Projects

Respiratory Sound Diagnostic Engine (CNN-based)

Dec 2025

- Engineered a DSP pipeline using Librosa to extract MFCC-based spectral features (6K+ dimensions) for medical classification
- Designed and trained a CNN (TensorFlow/Keras) achieving classification across 8 respiratory disease classes
- Integrated audio datasets with patient metadata using Pandas to improve feature representation and prediction context
- Developed an end-to-end inference system with Tkinter GUI for real-time audio input, prediction, and visualization
- Optimized model pipelines and serialization, reducing inference latency and improving deployment readiness

AI Query Router – Multi-Model LLM Gateway

Mar 2026

- Architected a distributed AI gateway using FastAPI to route queries across multiple LLMs based on complexity, latency, and cost
- Designed microservices architecture using Redis, Kafka, and PostgreSQL for scalable and high-performance query processing
- Implemented intelligent routing and multi-tier fallback (70B → 13B → 7B), ensuring 99.9% availability and reducing costs by 60%
- Built a Redis caching layer achieving 40–60% hit rate and reducing response latency to sub-20ms
- Developed event-driven logging pipeline using Kafka to track latency, usage, and cost metrics for real-time monitoring
- Containerized services using Docker, enabling modular architecture, scalability, and efficient deployment

Leadership

Technical Lead – Algorand Blockchain Club

June 2025 - Present

- Led 180+ members building blockchain applications and conducted Web3 and decentralized systems workshops

Technical Lead – Next Gen Innovators Club

July 2025 - Present

- Conducted 2 workshops on Git, GitHub, and end-to-end project deployment using Vercel, Render, and AWS

Education

Vignan Institute of Technology & Science

July 2024 – Present

B.Tech in Information Technology CGPA: 8.53/10

Hyderabad, Telangana

Technical Skills

Languages: Python, Java, TypeScript, JavaScript

AI Engineering: LLM Systems, Agentic AI, LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering

Backend & APIs: FastAPI, REST APIs, Microservices Architecture, Async Programming

Cloud & DevOps: AWS (EC2, S3, Lambda), Serverless Architecture, Docker, CI/CD Pipelines

Distributed Systems: Apache Kafka, Redis, WebSockets

Databases: PostgreSQL, MongoDB

Tools & Platforms: Git, Github, Docker, Postman, VS Code

Achievements / Certifications

- Winner: Hack With AI Hackathon (KL University 2026)
- Ranked among Top 4 teams at Vignan Hackathon 2025
- AI Engineer Certification (Udemy) – Data Science, Machine Learning, Deep Learning & Agentic AI
- Scalar Certified: Java Mastering the fundamentals

Data Structures & Algorithms

- Solved 300+ problems across LeetCode and GeeksforGeeks with consistent practice
- Strong in Graphs, Dynamic Programming, Trees, Greedy, and Arrays with deep problem-solving intuition
- Proficient in writing optimized solutions with rigorous time and space complexity analysis